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ARTIFICIAL INTELLIGENCE-BASED AGENT AND FRAMEWORK FOR CONTEXTUALIZED, PRIVATE AND DOMAIN-SPECIFIC OUTPUT DRIVEN BY USER-SPECIFIED CONTENT

Abstract

A framework provides an approach for utilizing contextualized content from a user's designed set of documents and private data sets to generate customized, contextualized, private, and domain-specific outputs of agents within an artificial intelligence computing environment and a supporting architecture. The agents and artificial intelligence computing environment include augmenting a language model with the contextualized content, and prompting the language model to generate defined, domain-specific outputs. Such agents enable computing systems to execute specific actions identified by a user that are external to the supporting architecture from the defined, domain-specific outputs.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION(S) [0001] This patent application claims priority to U.S. provisional patent application 63/533,607, filed on Aug. 18, 2023, the contents of which are incorporated in its entirety herein. In accordance with 37 C.F.R. § 1.76, a claim of priority is included in an Application Data Sheet filed concurrently herewith.

FIELD OF THE INVENTION

[0002] The present invention relates to the field of artificial intelligence. More specifically, the present invention is a framework for deployment of privately-hosted artificial intelligence models, in which contextualized information from user-specified content augments language models to enable customized actions as domain-specific outputs of artificial intelligence-based agents that leverage such augmented models.

BACKGROUND OF THE INVENTION

[0003] Existing technology for identifying content, and using such content for analysis and augmentation of private data sets encompassed by all of one's universe of documents and files regardless of platform or format, is quite limited. Such existing technology is confined to basic search functions that at best provide a list of files in response to a search, and does not allow for customized identification and extraction of user-specified content from such private data sets.

[0004] Artificial intelligence offers some promise for improvements in understanding and augmenting the contents of such data sets. Transformer-based models such as large language models are an artificial intelligence tool that have recently developed and become prominent; however, such models require large-scale, expensive and resource-consuming computing systems for their implementation, and involve highly-customized and complicated builds to be leveraged as base models by follow-on applications. To date, no enterprise-quality applications using such models for analyzing and augmenting a universe of files within a smaller privately-hosted environment have been produced, and there are no such applications that focus on private data sets from user-designated files for driving customized, domain-specific outputs.

[0005] Still further, use of such models to enable multi-agent, artificial intelligence-based processing within native operating systems have not been developed. There is therefore a need in the existing art for an approach in which an artificial intelligence-based agent is able to operate within one's universe of files and documents, and provide contextualized, dynamic, private and persistent outputs from private data sets within such files and documents. There is a further need in the existing art for a multi-agent approach that is able to operate within smaller computing environments, and across multiple operating systems, to provide deep, artificial intelligence-based analysis and augmentation of private data sets for such contextualized, dynamic, private and persistent outputs.

BRIEF SUMMARY OF THE INVENTION

[0006] The present invention provides a framework for enabling artificial intelligence-based agents, and more specifically agentic systems that are built on top of, and leverage, other complex machine learning tools and artificial intelligence models, to drive customized, contextualized, private, and domain-specific outputs. The framework of the present invention is, in one aspect thereof, an approach for analyzing content within a user's designated set of documents and files

through applications of machine learning tools and artificial intelligence models, where contextualized content identified in private data sets in those documents and files is used is augmented by language models, deterministic models, and long-term knowledge graphs, in a privately-hosted environment. This approach is embodied in a framework designed, in a general sense, to enable observability and data analytics for dynamic information discovery. The framework is supported by a software-based data architecture and platform that supports one or more artificial intelligence-based agents to provide semantic, natural language-based functions for searching, contextualizing, analyzing, and executing actions based on the information in private data sets in the user's documents and files to generate defined, domain-specific outputs.

[0007] The agents of the framework of the present invention enable computers to execute domain-specific tasks by becoming, in one sense, special-purpose machines that are much more personal to the user, allowing the user to accomplish tasks and actions either with the special-purpose machine itself, or using tools to connect with humans, external systems and/or other machines. The agents of the present invention allow any user to write software for computing systems in manner that allows users to express themselves conceptually, to vastly improve the accessibility and power of both computing systems, and machine learning tools and artificial intelligence models that are becoming increasingly ubiquitous with such accessibility and power.

[0008] Still further, the artificial intelligence-based agents of this framework provide a persistent curiosity interface for obtaining the customized, contextualized, private, and domain-specific outputs, and for enabling analytical functions of the user's private data sets. The persistent nature of the curiosity interface allows artificial intelligence-based agents with this framework to remember conversations, queries, searches, and outcomes thereof. The artificial intelligence-based agents and supporting data architecture and platform enable contextual augmentation of large amounts of disparate types of data, regardless of format, type, or particular application.

[0009] Objects, embodiments, features, and advantages of the present invention and its embodiments will become apparent from the following description of the embodiments, taken together with the accompanying drawings, which illustrate, by way of example, principles of the invention.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0010] The accompanying drawings, diagrams, graphs, and charts, which are incorporated in and constitute a part of this specification, illustrate several embodiments of the invention and together with the description, serve to explain the principles of the invention.

[0011] FIG. 1 is a system diagram illustrating elements of an artificial intelligence-based platform supporting an agentic operating environment, according to one aspect of the present invention;

[0012] FIG. 2 attached is an exemplary workflow for deploying an artificial intelligence-based agent, according to one embodiment of the present invention;

[0013] FIG. 3 is a flow chart illustrating steps in a process for deploying an artificial intelligence-based agent, according to a further embodiment of the present invention; and

[0014] FIG. 4 is a data flow diagram for a framework for providing contextualized content for an augmented language model from private data sets according to one embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0015] In the following description of the present invention, reference is made to the exemplary embodiments illustrating the principles of the present invention and how it is practiced. Other embodiments will be utilized to practice the present invention and structural and functional changes will be made thereto without departing from the scope of the present invention.

[0016] FIG. 1 is a system diagram illustrating elements of a framework **100** for artificial intelligence-based agents **150** according to one aspect of the present invention. The framework **100** includes a data architecture platform **140** that supports an agentic artificial intelligence operating environment in which private data sets among user-designated files **111** are analyzed to drive defined, domain-specific outputs **124** that are highly customized to the private data sets and users **120**.

[0017] The framework **100** enables analysis of such private data sets to generate contextualized content **160** from within the user-designed files **111** to execute specific actions from the defined, domain-specific outputs **124**. These defined, domain-specific outputs **124** are output data **180** in customized, contextualized, and private outcomes of the artificial intelligence-based agents **150**. These artificial intelligence-based agents **150** are built on top of machine learning tools **142** and artificial intelligence models **144** such as transformer-based models, which include language models (such as large language models, or LLMs, and “small” language models). Distinctions between sizes of language models **144** are effectively ones of training data set size and of the number of parameters to train models and generate results from natural language queries. Regardless, and for ease of convenience, in the present specification, both large and small language models shall be referred to as language models **144**, and it is to be understood that neither the claims nor the disclosure presented herewith shall be limited to a language model **144** of any particular size or type.

[0018] Regardless, such defined, domain-specific outputs **124** leverage the content of the sets of information **138** within the user-designed files **111** to accomplish particular domain-specific outputs **124** such as exemplary outcomes **190** of output data **180**. The artificial intelligence-based agents **150** of the framework **100**, as noted above, are supported by an underlying data architecture and platform **140**, as well as in accordance with one or more interfaces **170** for both users **120** and external systems **195** that integrate with the framework **100**. The framework **100** and underlying data architecture platform **140** enables multiple artificial intelligence-based agents **150**, which may operate either on localized devices or systems, or within cloud computing systems, and within native or isolated operating environments, to deliver any type of contextualized private and domain-specific outcome as output data **180**.

[0019] The framework **100** may also include one or more application programming interfaces (APIs) **172** that enable various activities of data processing elements **132**, including the artificial intelligence-based agents **150**. These APIs **172** provide interfaces **170** between various data processing elements **132** within the framework **100**, as well as for communicating with external systems **195**. For example, artificial intelligence-based agents **150** may integrate with internal APIs **172** at least for intake of sets of information **138** from the data lake **137**, populating data collections comprising the knowledge base, and integrating with knowledge graphs, and for certain output functions such as for example generating a transcription of a particular user-designated file **111**. Examples of where APIs **172** may be leveraged for communicating with external systems including creating and delivering a short as a document and file **191** to an external system **195** or platform, adding information to a customer relationship management system **196** or database, and sending an instruction **194** to actuate **193** a mechanical system such as robotic equipment. APIs **172** may be native to the framework **100**, or in conjunction with external systems **195**.

[0020] The framework **100** of the present invention is embodied within one or more systems and/or methods that are performed in a plurality of data processing modules **132** that are components within a computing environment **130** that also includes one or more processors **134** and a plurality of software and hardware components. These data processing modules **132** may be configured to run within external cloud computing environments (and accessed therefrom by the framework **100**), and also may be configured to run locally on devices hosting the framework **100**, such as on mobile computing devices, “smart” phones, earphones or earbuds, on other wearable, internet-enabled devices such as watches and eyeglasses, and in automotive platforms. Still further, one or

more of the data processing modules **132** may be configured to run within, and executed on, edge computing environments and be responsive to natural language instructions, either verbal, written, or gesture-based. The one or more processors **134** and plurality of software and hardware components are configured to execute program instructions or routines to perform the elements, modules, components, and functions described herein that together comprise and are embodied within the plurality of data processing modules **132**. The words “module” and “modules” as used herein, may refer to (and the data processing modules **132** may themselves comprise, at least in part) logic embodied in hardware or firmware, or to a collection of software instructions, written in a programming language, such as, for example, Java, Python, C, or assembly. One or more software instructions for such modules **132** may be embedded in firmware. It will be appreciated that the functional data processing modules **132** may include connected logic modules, such as gates and flip-flops, and may include programmable modules, such as programmable gate arrays or processors. The data processing modules **132** described herein may be implemented as either software and/or hardware modules and may be stored in a storage device. It is to be additionally understood that the data processing modules **132**, and the respective components of the present invention that together comprise the specifically-configured elements, may interchangeably be referred to as “components,” “modules,” “algorithms” (where appropriate), “engines,” “networks,” and any other similar term that is intended to indicate an element for carrying out a specific data processing function. The framework **100**, and the artificial intelligence-based agents **150**, may enable such a computing environment **130** to effect one or more special-purpose machines on which such a computing environment **130** is configured, and may further enable machines on which such a computing environment **130** is configured to themselves act as special-purpose machines as noted above.

[0021] At least the supporting data architecture platform **140**, machine learning tools **142**, and artificial intelligence models **144** are all part of a broader machine learning and artificial intelligence environment within the framework **100**. The implementations and applications of this broader machine learning and artificial intelligence environment within the framework **100** enable the processing of private user data in the input data **110** and drive the defined, domain-specific outputs **124** of the framework **100**, as well as output data **180** and exemplary outcomes **190** that embody the defined, domain-specific outputs **124**. This broader machine learning and artificial intelligence environment enables the artificial intelligence-based agents **150** to further enable computers and computing systems with which the artificial intelligence-based agents **150** are configured to execute domain-specific tasks, and therefore transform those computers and computing systems into special-purpose machines that are much more user-effective by accomplishing tasks and actions either with the special-purpose machine itself, or in conjunction with external systems **195** and/or other machines.

[0022] The framework **100** includes multiple implementations of machine learning tools **142**. These machine learning tools **142** may include many different techniques of machine learning, such as both supervised and unsupervised learning as well as instantiations of neural networks to continually enhance the data processing functions performed in the framework **100** of the present invention, by developing and understanding relationships between various types of information. The framework **100** also includes multiple additional artificial intelligence (AI) models, which may include standardized models or models developed for particular purposes, such as neural-network-based transformer models that include language models **144**, including large language models (LLMs), and may also include one or more models customized according to proprietary formulas. Regardless, the machine learning tools **142** and language models **144** are comprised, at least in part, of algorithms that apply many different mathematical approaches to analyzing information (either structured or unstructured) and generating the defined, domain-specific outputs **124** that provide the customized, contextualized, and private outcomes of artificial intelligence-based agents **150** within the framework **100**.

[0023] Language models **144** are programs that are able to recognize and generate natural language in text, among other tasks. Large versions of these language models **144** are trained on huge sets of data. Regardless of size (small or large), language models are built using machine learning techniques, such types of neural networks commonly referred to as transformer models. These neural networks include implementations of deep learning techniques for understanding natural language inputs and how characters, words, and sentences function together. Deep learning involves the probabilistic analysis of unstructured data, which eventually enables the neural network to recognize distinctions between pieces of content without human intervention.

[0024] Returning to FIG. **1**, the framework **100** ingests, receives, requests, or otherwise obtains input data **110** from many different types of sources. Input data **110** may include many different types and formats of information that form user-designed files **111**. Input data **110** may include documents **112**, images **113**, audio files **114**, video files **115**, and messages **116**. Users **120** may also designate input data **110** that includes Internet addresses **117**, web links **118**, and any other type or format of private user data **119**. Input data **110** may also include external data **122** from external sources. Input data **110** may be obtained from any source, and be in any format.

[0025] Examples of user-designated files **111** may include, but are not limited to, written documents such as those created or maintained with document creation and management systems such as Microsoft Office™ and Drive™ or Docs™ provided by Google™. Messages **116** may include those created and maintained using messaging systems such as for example Outlook™, Gmail™, Slack™, WhatsApp™, and Viber™ (including, but not limited to, text messages and emails via any platform or provider). User-designated files **111** may further include any other electronic file containing decipherable content (such as for example PDFs, JPEGs, etc.), regardless of whether such files are comprised of structured data, unstructured data, or both. Such user-designated files **111** may also include files collected from virtual meetings platforms in the form of meeting recordings (either audio or video or both), and transcriptions thereof.

[0026] The input data **110** may include information provided by the user **120** in the form of queries, instructions, or other approaches to entering information within the framework **100**. This type of input data **100** may be verbal, written, or gestured; it may also be determined from real-time interactions or time-lagged interactions, such as for example verbalized or written instructions during a recorded conversation, which occur in real-time or in a prior recording thereof.

[0027] The input data **110** may still further include information provided by a user **120** that provides definition for the defined, domain-specific output **124**. Such information may be provided via one or more prompts or sub-prompts **164** of an artificial intelligence-based agent **150**. This information includes the domain itself—the subject matter for which the artificial intelligence-based **150** is being created to generate the output **124** desired. Domain specificity in the output context may be any type of output **124** desired by the user **120**, and may be thought of as the reason for the artificial intelligence-based agent **150**. Domains may therefore represent a subject or theme of the artificial intelligence-based agent **150**, as well as the subject of its outcome. Specificity of the domain may therefore be constraints or parameters for the domain, and constraints or parameters for the defined, domain-specific output **124** itself. For example, the framework **100** may be implemented to build a ranking agent (i.e. enrollments, mental health risk); an accessibility agent (i.e. notes, summaries); a knowledge base (i.e. patent database questions); shorts (i.e. highlights short video or audio); an assessment agent (i.e. assessment without ranking); a notification agent (i.e. meeting prep email); an agent that generate one or more forms (i.e. RFC response, or a request to perform maintenance on equipment or devices); and an auto-grader agent (i.e. assessment a student's performance on a test). Each of these artificial intelligence-based agents **150** has one or more domain constraints, such as its field, what its function(s) are, what information is to be analyzed, and what its outcome is to be. These domain constraints may be provided to the artificial intelligence-based agent **150** as natural language prompts **164** (either by textual, verbal, or gesture-based entry). Therefore, prompts **164** may be considered as one form of input data **110** for the

framework **100**. Other ways of introducing domain-specificity may include analysis of the user-designated files **111** by the artificial intelligence-based agent **150**; in other words, the artificial intelligence-based agent **150** may learn from prompts **164** while processing user-designated files **111** to create domain-specificity, depending on the circumstances for which it is being asked to analyzed such user-designated files **111**.

[0028] All of these are types of artificial intelligence-based agents **150** and represent applications and use cases **197** of the framework **100**, and it is to be understood that many such applications and use cases **197** are possible and within the scope of the present invention. It is to be further understood that neither the claims herein or the present disclosure are to be limited to any one type of artificial intelligence-based agent **150** or specific application or use case **197** discussed herein.

[0029] Domain specificity may also be temporal, in addition to contextual, thematic, and subject driven. For example, an artificial intelligence-based agent **150** may be asked to analyze a particular data set over a specific period of time (for example, last month, last week, last period, last quarter, etc.). Therefore, temporality may also be introduced as a domain constraint of a defined, domain-specific output **124**.

[0030] It is contemplated that a language model **144** selected for application within an artificial intelligence-based agent **150** may itself be domain specific. In other words, a particular language model **144** may be small and trained on data that is within a specific domain. Therefore, an additional layer of domain specificity may be introduced by the selection of a particular language model(s) **144**.

[0031] Regardless of type of input data **110** or the form of a user-designed file **111** (or type of external data **122** or defined, domain-specific output **124**), information is ingested into the framework **100** by a data intake element **136**. This data intake element **136** enables formation of a data lake **137**, within which at least the sets of information **138** contained within the user-designed files **111** are maintained and stored, including that which represents the defined, domain-specific output **124**.

[0032] Prompts **164** may include natural language instructions that are either textual, verbal, or gesture-driven. Prompts **164** may include nested or sub-prompts, and may be provided by a user **120** directly, or may be created by the artificial intelligence-based agent **150** in response to earlier prompts **164** (for example, in the case of nested or sub-prompts). In addition, prompts **164** may be pre-prepared and made available to the user **120** such no textual, verbal, or gestured instruction need be given. The framework **100** may include a roster or set of specific actions **192** that are the result of prompts **164** for defined, domain-specific outputs **124** (such as for example write to customer relationship management system **196**, create PDF, send email, grade questions, etc.), and indicia for performing these types of specific actions **192** are available to the user **120**, for example as a radio button on interface **170**. The user **120** may be able to chain such specific actions **192** together to create a very specific task (or, a very specific defined, domain-specific output **124**). In this manner, the user **120** may be able to create prompts **164** without entry of anything at all other than selecting indicia for specific actions **192**.

[0033] The framework **100** may further include a specific artificial intelligence toolkit that exposes these specific actions **192** along with precise descriptions, or prompts **164**, to an augmented language model **162**. These prompts **164** for the augmented language model **162** enable the augmented language model **162** to select which of these specific actions **192** are appropriate and then execute those specific actions **192**. This paradigm, and this specific artificial intelligence toolkit, enable the framework **100** and an artificial intelligence-based agent **150**, to present very specific outcome sets for users **120**.

[0034] Prompts **164** within the framework **100** may therefore present distinct phases for users **120** within artificial intelligence-based agents **150**. In one such phase, the framework **100** presents curated specific actions **192**, and allow users **120** to provide artificial intelligence-based agents **150** with domain-specific knowledge, either through prompts **164** or through another interface **170** for

providing such knowledge for a defined, domain-specific output **124**. In a further phase, the framework **100** allows users **120** to curate their own specific actions **192** and provide prompts **164** within specified domains. And in still a further phase, the framework **100** provides artificial intelligence-based agents **150** in which an augmented language model **162** decides on specific actions **192** based on prompts **164** from users **120** for dynamic tasks that are in addition to the static and repeatable tasks in the defined, domain-specific outputs **124**. This enables creation of nested specific actions **192** that are chainable, such that with every building block action created, the artificial intelligence-based agent **150** adds an additional n number of possible interactions and chains.

[0035] The supporting data architecture platform **140** of the framework **100** of the present invention may also leverage a retrieval augmented generation (RAG) architecture **146** to analyze structured, unstructured, and temporal data within the user-designated files **111**. Retrieval-augmented generation (RAG) is a process of optimizing the output of a language model **144** that enables extension to specific domains or to a private data sets among user-designated files **111**, without having to retrain a model that has been augmented.

[0036] Without RAG, a language model **144** takes user data and creates a response based on information that it already knows. With the RAG architecture **146**, the framework **100** is able to extract relevant information from structured data portions of private data sets in the user-designated files **111** to create additional contextualized content **160** that may then be added to the augmented language model **162**. The RAG architecture **146** enables numerical information—in both structured and unstructured portions of private data sets—to be included in the contextualized content **160** within the framework **100**, providing a richer set of information **138** for the augmented language model **164**. The RAG architecture **146** also further enables, as noted above, the augmented language model **164** to remain privately-hosted in that it does not need to be re-trained external to the artificial intelligence-based agent **150** or framework **100**.

[0037] The application of a RAG architecture **146** to the private data sets of users **120** also enables the artificial intelligence-based agents **150** to reduce the possibility of hallucinations or incorrect answers to prompts **164**. Regardless of whether directed at structured data, unstructured data, or combinations of both in such private data sets, the application of a RAG architecture **146** improves the outcomes generated in defined, domain-specific outputs **124**.

[0038] An effective RAG application enables location of information relevant to the user's prompt **164** and supplies that information to the augmented language model **162**. As noted herein, the framework **100** identifies text relevant to the user's prompt extracting the text therein in a way that preserves structured information such as tables of data; the final text is chunked into snippets of text, and an embedding vector for a language model **144** is calculated for each document chunk and stored. These embeddings are later accessed to easily recall snippets from documents that are relevant to prompts **164**.

[0039] While transformer models (such as large language models) perform very well at providing coherent and thorough responses to user prompts, they are also suffer from reliability issues. This is because LLMs often respond with incorrect or fabricated answers, known as hallucinations, This occurs because an LLM, by itself, does not know anything that happened after it has been trained, and importantly in the context of the present invention, lacks access to information in private data sets. On a general level, attempts to address this issue may involve providing the LLM with the information it needs to address the user's prompt; but current approaches to doing this are only able to provide the LLM with just a few pages of documents, and therefore very limited. But where large scale private data sets in a plurality of user-designated files **111** are being analyzed, the RAG architecture **146** of the framework **100**, when applied thereto, provides an augmented language model **164** with additional information to reduce the possibility of these hallucinations.

[0040] Implementation of a RAG architecture **146** with private data sets in user-designated files **111** reductions in hallucinations and incorrect responses to prompts **164** in a number of ways. For

example, implementation of a retrieval component such as RAG architecture **146** that searches through private or domain-specific datasets provides access to specific knowledge that allows the augmented language model **162** to pull in more accurate and contextually relevant information for both the prompt **164** and the defined, domain-specific output **124**. This means that instead of relying solely on potentially vague or generalized knowledge encoded with a language model's parameters, the augmented language model **162** has access to detailed, up-to-date, and domain-specific information directly from the set of information **138** contained within the private data set(s) of the user **120**.

[0041] The retrieval process in the use of RAG architecture **146** provides contextual relevance (domain-specificity) which helps ensure that the information the augmented language model **162** uses to generate responses is highly relevant to the specific query or context. This reduces the likelihood of generating responses based on incorrect or generalized data that might not fit the specific scenario being addressed.

[0042] Still further, use of a RAG architecture **146** in the framework **100** provides for a reduction in ambiguity. When the artificial intelligence-based agent **150** retrieves information from a curated dataset, it has access to more precise data, which helps clarify ambiguous or complex queries. This reduces the chance of the augmented language model **162** making up details or providing misleading answers.

[0043] An artificial intelligence-based agent **150** within the framework **100** may be instantiated and deployed, in an exemplary implementation thereof, according to a generalized workflow **200** according to the diagram of FIG. 2. In this workflow **200**, the framework **100** gathers data **210** for a specific agent (for example, a telephony agent). This data is then loaded **220** as a group of documents that are defined by a user at least according to the type of agent and its intended function(s).

[0044] The workflow **200** proceeds with creation **230** of an artificial intelligence-based agent. The creation **230** of an artificial intelligence-based agent **150** includes both identifying a specific context **240** for the artificial intelligence-based agent **150**, and define one or more responsive outputs **250**. Together, these aspects of the workflow **200**—identifying context **240** and defining a responsive output **250**—create the domain-specificity in the defined, domain-specific output **124** of the specific artificial intelligence-based agent **150**.

[0045] Identifying a specific context **240** for an artificial intelligence-based agent **150** may involve several issues. For example, the workflow **200** identifies documents **241** comprising the one or more user-designated files **111** that provide the information sources for the artificial intelligence-based agent **150**. Identifying a specific context **240** may also involve determining a protocol **242** for adding new documents to the context, as the specific agent **150** being created **230** may have documents added periodically or continually. Regardless, these documents may be ranked **243** according to their relation with each other, for example according to a cosine similarity. This allows for the group of documents **220** to be contextualized according to various characteristics, such as for example people, topics, entities, emotions, etc. Identifying a specific context **240** may also involve identifying a temporal context **245**. For example, this may involve identifying specific time periods (such as last month, last week, last period, last quarter, etc.).

[0046] Further, identifying a specific context **240** may also involve determining a toggle **246** for curated documents against all documents. This allows for the specific artificial intelligence-based agent **150** to be further contextualized based on a weighting of information among the group of documents **220**, so that curated documents (at least where one or both of topics **244** and temporality **245** have been contextualized, and documents have been ranked **243**) may be assigned greater authority for the defined responsive output **250** being modeled. Finally, the identifying a specific context **240** may include identifying, selecting, or assigning a language model **144** for the specific artificial intelligence-based agent **150** (or a combination of language models **144**). This is due to some language models **144** being more relevant to particular types of documents (or content

thereof) among the user-designed files **111**, and/or more relevant to the defined, domain-specific output **124** of the artificial intelligence-based agent **150**.

[0047] The workflow **200** also includes defining agent functions **252** for the responsive output **250** to achieve the defined, domain-specific output **124**. Agent functions define the steps involved in performing the defined, domain-specific output **124**. This may include code for executing specific actions **192** or any other type of responsive output **250** of the one or more prompts **164** that are applied to augmented models **162** in the framework **100**, and may also include protocols for interacting with external systems **195**, such as for example actuating external devices **193**. It is to be understood that many exemplary outcomes **190** embodying such defined, domain-specific outputs **124** are possible and within the scope of the present invention, and therefore neither the claims nor this disclosure are to be limited to any specific example of either a responsive output **250** or defined, domain-specific output **124** of an artificial intelligence-based agent **150** discussed herein. It is to be further understood that an artificial intelligence-based agent **150** according to the present invention may have any type of defined, domain-specific output **124**, and that workflow **200** enables definition of any type of responsive output **250** to achieve such a defined, domain-specific output **124**.

[0048] The workflow **200** may also include creating agent features **260**. Each artificial intelligence-based agent **150** may have different features **260**, for example depending on the defined agent functions **150** and the defined, domain-specific output **124**. These features **260** may assigned within each artificial intelligence-based agent **150** according to one or many subscription levels **262**.

[0049] The workflow **200** may also include defining approaches to interaction **270** with each artificial intelligence-based agent **150**. Each such artificial intelligence-based agent **150** may have one or more interfaces **170** through which they may be interacted with. For example, an artificial intelligence-based agent **150** may be configured with built-in, custom prompts **272**, for example as “radio” buttons on an interface **170**. A chat feature **274** may also be configured, such that a user **120** may converse with an artificial intelligence-based agent **150** via a chat function on the interface **170**. Still further, a user may interact **270** to generate **276** their own domain-specific output **124** via an interface **170**. This may occur, for example, through a feature on such an interface **170** that allows the user **120** to adjust agent options **278**. Adjusting agent options **278** may also include features that allow a user **120** to change a data toggle or choose a specific language model **144**.

[0050] The workflow **200** also enables an artificial intelligence-based agent **150** to be refined **280**. Refining **280** an artificial intelligence-based agent **150** includes adjusting context **282**—this may be accomplished by refining **280** the characteristics of a specific context **240** discussed above. Refining **280** an artificial intelligence-based agent **150** may also include altering **284** or adjusting the prompts **164** (and any nested or sub-prompts), as well as re-defining, changing, adding or deleting a domain-specific output **124** by re-defining a responsive output **250** within the workflow **200**.

[0051] FIG. **3** is a flow chart illustrating steps in a process **300** for deploying an artificial intelligence-based agent **150** according to another embodiment of the present invention. In the process **300** of FIG. **3**, input data **110** comprised of one or more user-designated files **111** is received at step **310** to analyze the set(s) of information **138** contained therein within an artificial intelligence-based agent **150**. Each artificial intelligence-based agent **150** as noted herein analyzes the input data **110** in one or more prompts **164** of an augmented language model **162**. The process **300** includes a mechanism that allows a user **120** to define a domain-specific output **124** of the artificial intelligence-based agent **150** and the supporting data architecture platform **140**. The domain-specific output **124** may be defined as part of the input data **110** provided by the user **120**, for example via an interface **170**.

[0052] At step **330**, the process **300** prepares contextualized content **160** from set(s) of information **138** in the user-designated files **111**. This is accomplished using machine learning tools **142** within the supporting data architecture platform **140**. The machine learning tools **142** extract data points

from unstructured information at step **340**, relative to the defined, domain-specific output **124**. At step **350**, the machine learning tools **142** are applied within the framework **100** and process **300** to create embeddings from extracted data points, and calculate embedding vectors that represent the contextualized content **160**.

[0053] The process **300** continues with augmenting a language model **162** at step **360** with the contextualized content **160**. The artificial intelligence-based agent **150** is thus ready for prompting at step **370**; in this step, the process **300** applies one or more prompts **164** within the artificial intelligence-based agent **150** to drive functions that compile and generate the defined, domain-specific output **124**.

[0054] At step **380**, the process **300** generates the defined, domain-specific output **124** from the artificial intelligence-based agent **150**, and delivers the defined, domain-specific output **124** in a manner previously-defined by the user **120**. At step **390** the process **300** may execute one or more specific actions **192** (or some other form of the defined, domain-specific output **124**). This may include executing a specific action **192** at or within an environment that is an external system **195** to the supporting data architecture platform **140**. This may also be previously defined by the user **120** as part of the defined, domain-specific output **124**.

[0055] The framework **100** may be implemented in many ways. In one embodiment of the present invention, the supporting data architecture platform **140** may be thought of as including both server-side components and client-side components for implementing the artificial intelligence-based agents **150**, and utilizing language models **162** for application within such artificial intelligence-based agents **150**, to generate the defined, domain-specific output **124** based on contextualized content **160** from designated user files **111**. The present invention may therefore be implemented in a manner that includes input data **110**, processing of such input data **110** in conjunction with language models **144** at least by analyzing instructions in one or more prompts **164**, and generating outputs **124** that represent domain-specific contextualizations of the user's data.

[0056] In one embodiment of the present invention, on a server side, the supporting data architecture platform **140** gathers the user-designated files **111** (such as documents, video files, audio files), and any other types of files desired by the user **120**, and regardless of where such files are stored or accessed from. These collective documents may be initially processed to perform initial functions that prepare them for subsequent functions or handling within the supporting data architecture platform **140** (such as transcription, diarization with speaker identification, and then summarization), and also infused with any other platform features selected by the user (for example, assigned with detected emotions). All files may then be structured with various general identifiers (date, type, user, document name, and document ID) and these general identifiers may be stored in a separate database table.

[0057] Continuing with this embodiment, one or more processors on the server side of the supporting data architecture platform **140** then begin compiling information such as document title and text chunk from each document in the user-designated files **111**, and create individual embeddings for each document based on the text chunk for the augmented language model **162**. A database may be utilized with these processors to govern when the processors perform these tasks for documents. Individual embeddings for text chunks are then used to obtain embedding vectors from the large language model, and then stored with a vector database to associate these embedding vectors with a text chunk and a document ID for each document. In a vector database, each entry may have a chunk (text), vector embedding and a document ID which it belongs to, along with a unique key for each entry.

[0058] Regardless, all of these tasks on the server side happen once per document; however, it is to be noted that whenever underlying text in the document is created or updated, the text needs to be re-chunkified and embeddings recalculated. Every time a document is edited, it is marked accordingly so that each embedding and chunk can also be updated. In effect this means that a

modified document goes through the pipeline described above again; alternatively, only a few chunks may be updated. Where APIs are utilized to communicate with language models **162**, the APIs transfer individual embeddings for text chunks when the language model **162** is accessed by a user **120** on the client side. To be more specific, when an artificial intelligence-based agent **150** is prompted (or when a user **120** poses a question), the supporting data architecture platform **140** of the present invention chunkifies the text, calculates embedding vectors for that prompt, then performs a similarity search within the vector database to find relevant embeddings to supply as context to the augmented language model **162**. This prompt, complete with contextualized context **160** representing sets of information **138** in the documents in the user-designated files **111**, is sent to the LLM via an API call.

[0059] A chunk is a section of the document which is converted into text format. Each chunk may be a specified number of words, but the chunk size will vary depending on the context and amount of information needed to be captured. Embeddings are a different representation of text, in mathematical form—a vector, which is an array of data representing the original chunk of text. Embeddings may be generated depending, at least in part, on the type of language model **144** being utilized. Embeddings are a list of numbers (vectors) extracted or derived from the text in the chunk. Each method of embedding has a set size, and is not affected by how many words are passed in.

[0060] An embedding is also a mapping of a discrete variable to a vector of continuous numbers. In neural networks, embeddings are continuous vector representations of these discrete variables. Neural network embeddings reduce the dimensionality of variables, and enable a meaningful representation of variable categories in the transformed space. They also enable the finding of nearest neighbors in the embedding space. Embeddings therefore may be also used for inputs to neural networks among the machine learning tools **142** for a supervised task that looks for similarities in the embedding space.

[0061] DynamoDB is one database approach that may be used to compile data and metadata produced at each step in the processing pipeline of the supporting data architecture platform **140** for the artificial intelligence-based agents **150**; it may also be used as a message broker to start the next processing step. Regardless of the approach used or specific type of environment in the supporting data architecture platform **140**, the processing pipeline takes user-designated files **111** (for example, mp4 recordings of meetings, podcasts, YouTube videos) and analyzes them, generating a handful of data and metadata therefrom (in transcriptions, diarization, etc.). The supporting data architecture platform **140** takes generated data and metadata from the user-designated files **111**, and creates a pipeline step that transcodes this information to a format that can be ingested by augmented language models **162** (text chunk embeddings). The supporting data architecture platform **140** then uses the embeddings for the set of information in the documents comprising the user-designated files **111** as the contextualized context **160** for the augmented language model **162** for and artificial intelligence-based agent **150**.

[0062] In a further embodiment of the present invention, the client side of the supporting data architecture platform **140** provides for real-time interaction between a user **120** and the artificial intelligence-based agent **150**, where such a feature is enabled (in effect, a chatbot feature). In such a feature, users **120** may pose their own prompts **164** of the artificial intelligence-based agent **150** to ask questions via a front-end interface **170**. The real-time interaction occurs using a native application or other such interface **170** that supports entry of prompts for the artificial intelligence-based agent **150**.

[0063] In this further embodiment, the supporting data architecture platform **140** creates a further embedding from the question posed by the user **120**. The supporting data architecture platform **140** then filters documents from the user-designated files **111** based on the user's access to only that user's available media library (and whatever other information the user designates and has access to, to form his or her universe of documents to be analyzed).

[0064] The supporting data architecture platform **140** then performs an index lookup on filtered

embeddings in the vector database and retrieves plain text chunks based on an embeddings similarity. The index lookup may be a form of database search, in which a nearest neighbor approach is used to analyze chunks that are closest to what has been asked of the artificial intelligence-based agent **150**. This is in effect a similarity search between one embedding vector and a data store of other embedding vectors. To further clarify with an example, consider an engine for analyzing telephony data: give it a voice recording of a telephone conversation with parameters for how to analyze the telephone conversation, and it generates an embedding vector which is a representation describing portions of that voice recording according to those parameters. It then searches for similar portions of other voice recordings in a vector datastore and returns results that are highly similar. Such an engine therefore enables comparisons of telephone conversations across time, which further enables analytics based on those telephone conversations.

[0065] The embeddings similarity may be performed using a cosine similarity, which is a mathematical implementation of an approach such as nearest neighbor that determines how close the vectors (embeddings) are to each other. The supporting data architecture platform **140** compares the question embeddings with all the filtered chunks/documents to get the most relevant list of chunks. Every embedding is also a vector by itself, represented as a series of numbers.

[0066] The original prompt **164** posed by the user **120** is now contextualized with supplemental data for further processing of the prompt **164**. The supplemental data that contextualizes the prompt **164** represents the plain text chunks based on the embeddings similarity. The prompt **164** is then passed together with the chunks to the augmented language model **162**. The answer is retrieved either provided to the user **120** (on an interface **170**), or compiled as part of the defined, domain-specific output **124**.

[0067] The conversation or prompt **164** history may also be saved in a backend database of the supporting data architecture platform **140**, together with answer pairs and other information such as timestamps. In this embodiment, every time a set number of new prompts **164** are presented, that history is also passed to that portion of the supporting data architecture platform **140** discussed above responsible for creating embeddings from the prompt **164**. In this manner, the new embeddings created from the new question can be additionally contextualized with metadata representing a conversation history saved from the prior interactions with the augmented language model **162**. This enables the artificial intelligence-based agent **150** to be persistent, remembering the user's history of prompts **164** and outcomes of searches initiated by those prompts **164**.

[0068] The threshold for the number of queries posed in a prompt **164** asked before a history is passed to the supporting data architecture platform **140** may change, depending at least upon the flexibility of the artificial intelligence-based agent **150** and ability to learn over time, as well as the number of tokens need for the augmented language model **162** (which relates to the amount of information that can be passed through the API).

[0069] The supporting data architecture platform **140**, and also the entire framework **100** of the present invention, ensures that no confidential information from the user **120** ever leaves their system, and therefore no confidential information presented in a prompt **164** using the artificial intelligence-based agent **150** will be passed to the language model **144** itself. This is true at least in part because of the privately-hosted nature of the augmented language model **162**, and selection of the language model **144** itself that is used, as the present invention enables selection of particular language models **144** which enable hosting in a private, secure operating environment.

[0070] The choice of language model **144** also influences the underlying processing hardware that is required. The framework **100** of the present also allows for language models **144** to be selected where CPU-based processing is possible, rather than GPU-based processing. This enables a fully localized approach, where the private hosting of the language model **144** occurs entirely within a local device using on-board CPUs. This further ensures that no confidential information leaves the user's system. It also ensures that native artificial intelligence-based agents **150** may be developed on such devices, and using local device-specific operating systems, such that each artificial

intelligence-based agent **150** may implement its own privately-hosted language model environment for processing information such as prompts **164**.

[0071] The supporting data architecture platform **140** and framework **100** also enable document provenance within the user-designated files **111**. An embeddings search returns similar embeddings relevant to a search query presented by a prompt **164**, which is then linked to particular documents. From the supporting data architecture platform **140**, the framework **100** may not only provide an answer to a prompt **164**, but also provide a list of documents that directly relate to a question-answer pair. A user **120** may thus look through relevant documents as provenance for the answer to the prompt **164** they have made to the artificial intelligence-based agent **150**.

[0072] Still further, the supporting data architecture platform **140** enables natural language-based document semantic search, which directly relates to such document provenance. For example, the interface **170** incorporating the chatbot function of the artificial intelligence-based agent **150** may include a search bar; the user **120** writes in the desired search terms, and because the present invention enables document provenance, the chatbot returns all documents relevant to the search.

[0073] FIG. **4** is a data flow diagram for an artificial intelligence-based agent **150**, according to one embodiment of the present invention. FIG. **4** illustrates a process **400** for internal processing of user-designated files **111** in response to prompts **164**.

[0074] In this process **400**, a user-centric document store **410** represents a data lake **137** for the user-designated files **111** for a particular user **120** of an artificial intelligence-based agent **150**. The user-centric document store **410** is the set of documents that linked to a user **120** account, and includes data packages **412** used by a particular artificial intelligence-based agent **150** for that user-centric document store **410** (such as for example user-uploaded documents and media **413** (corporate policy documents, presentations, recordings, pdfs), and user meeting records **414** (collected by a bot that attends virtual meetings on a user's behalf. If the document in the user-designated files **111** in the user-centric document store **410** is a media file, it is transcribed and diarized into text. Alternatively if a document is for example a Portable Document Format (PDF) file, the framework **100** extracts the text therein in a way that preserves structured information such as tables of data. Regardless of source, the final text is chunked into snippets of text, and an embedding vector for a language model **144** is calculated for each document chunk and stored in a vector database **415**. These embeddings are used later on to easily recall snippets from documents that are relevant to prompts **164**. The framework **100** also stores metadata (e.g., meeting time, relevant people, etc.) for each document in a metadata database **416**.

[0075] The process **400** then performs a prompt analysis **420**. A user **120** enters information as a chat, or starts a task, via the artificial intelligence-based agent **150**, which begins by issuing a prompt **164**. In this embodiment, for each of the below steps in the prompt analysis **420**, the framework **100** issues an internal prompt based on the original prompt **164** to extract information about the original prompt **164** and operate in different ways depending on what the artificial intelligence-based agent **150** determines about the query or task being performed. The artificial intelligence-based **150** therefore asks a series of questions about the original prompt **164**: for example, what time period are you asking about? are you asking for specific facts or wanting me to summarize some information, or something else?

[0076] The framework **100** performs a filter extraction **421** on the prompt **164**. The framework **100** extracts any filters **422** from the prompt **164** to curtail the document search space when gathering supporting data to provide context for the query represented by the prompt **164**. The framework **100** looks for entities the user **120** might know about in their related knowledge graph, such as people, organizations, and tags. The framework **100** also looks for temporality in mentioned date ranges or periods, such as “last quarter” or “in the last week”. Whenever the framework **100** goes to look up data from documents to answer a query in a prompt **164**, these filters **422** are first applied to the search space, for both performance as well as quality of results returned. Filters **422** enable filtered documents **411** for the user-centric document store **410**.

[0077] The process **400** then strips **423** filters **422** identified in the filter extraction **421** from the prompt **164**. Once the artificial intelligence-based agent **150** knows the filters **422** to apply to the search space, it performs this strip **423** function to remove this information out of the original prompt **164**, and the artificial intelligence-based agent **150** accomplishes this by running another prompt. For example, the query “what issues has my sales team been experiencing in the last quarter?” would result in a date filter of “last quarter” and the resulting post-processed prompt **424** for the artificial intelligence-based agent **150** with the filter **422** stripped would be “what issues has my sales team been experiencing?”.

[0078] The artificial intelligence-based agent **150** then performs a prompt type extraction **425** to determine the type **426** of the prompt **164** by classifying it into a few different buckets (multi-class classification): recall **431**, insight **433**, metadata **438**, and more (to create contextualized data relevant to the prompt **164**). Each type **426** of prompt **164** proceeds along its own processing flow which changes: (i) how it grabs data, (ii) what data it grabs, and (iii) how it processes this data. Each of these flows forms a high-quality set of data to support the user's original prompt **164** to answer queries in such a prompt **164** or to accomplish tasks requested of the artificial intelligence-based agent **150**. This also provides the foundation of a high-quality RAG architecture **146**, by injecting the right data to support the prompt **164**.

[0079] The process **400** then proceeds with supporting data composition **430**. The framework **100** applies this supporting data composition **430** step to get the best supporting data possible to provide alongside the original prompt **164** for the artificial intelligence-based agent **150**. Based on the prompt type **426** extracted in the previous step, the framework **100** may operate differently.

[0080] In recall **431**, type classification for this supporting data composition **430** step generally does not require extensive processing. Recall **431** is utilized to look up key details from a pool of documents or data that are relevant to the prompt **164**, so that the artificial intelligence-based agent **150** can either make inferences about the recalled data or regurgitate it to answer a query. For example, a **150** page corporate annual report is upload, and the artificial intelligence-based agent **150** is prompted with “what percent of gross revenue came from sales of product y” (y being a product of the corporation). The framework **100** calculates the embedding vector for the query in the prompt **164**, and then searches through all the precalculated embedding vectors in the vector database **415** for the text in the annual report to find text snippets with high cosine similarity **417**. These chunks of text would indicate portions of the document where revenue from product y were mentioned so that the query may be answered.

[0081] Consider a further example prompt **164**: “What are our end of year sales targets for agents?”. This is a straightforward classification of the prompt type **426** from the immediate, basic parameters of the prompt **164** (as is the example in the paragraph above). The framework **100** proceeds calculates an embedding vector for the prompt **164**, and then runs a cosine similarity **417** across the vector database **415** to get the top x document text snippets **432** (after filtering) to answer the query in the prompt **164**.

[0082] Insight **433** for type classification in the supporting data composition **430** step requires additional analysis due to the more complex nature of the prompt **164**. Consider the example prompt **164**: “Summarize this podcast about climate change and write a haiku about the author's stance on fossil fuels”. If the podcast is very short and the text fits in a context window for the augmented language model **162** being used, then classification of the prompt type requires no further processing. If however the podcast extensive—for example, it is an hour long with a long transcript—this information is compressed to fit into the context window **434** of the augmented language model **162**. The framework **100** chunks the document text into chunks as normal, and calculates embedding vectors for each chunk, but then clusters **435** these embeddings. This is because each natural cluster represents a core idea about the document. The framework **100** calculates the centroid of each cluster **435** and then from this centroid, computes the Euclidean distance with every embedding vector in the cluster **435** and chooses the top x embeddings. This

effectively compresses the text of the document while still maintaining important concepts and ideas explored in the document. Then the framework **100** checks if it is within the context window of the augmented language model **162**. If it is still beyond this context window, the framework **100** iteratively prunes **436** the text using cosine similarity of each embedding against the embedding vector of the input prompt **164**. The lowest similarity document chunk is discarded, and so on, until the final supporting data can fit within the available context window of the language model **144** alongside the prompt **164**. This creates the contextual data **437** for the prompt **164**.

[0083] For classification of metadata **438**, consider the example prompt: “When was my last meeting with Jim?”. These types **426** of prompts **164** are queries regarding the data state of the underlying data architecture platform **140** or metadata of documents the user **120** has access to in the underlying data architecture platform **140**. For these queries the artificial intelligence-based agent **150** references the relational database of document metadata (the metadata database **416**) consisting of structured and unstructured data of relevant documents **439** with metadata records to inject alongside the prompt **164**. This includes items such as recent meetings, documents, uploads along with their metadata such as participants, important dates, tags, etc.

[0084] In prompt formation **440**, the underlying data architecture platform **140** constructs a post-processed prompt **424** for the augmented language model **162**. The framework **100** first gathers any previous prompt, and response pairs in the user's session or task action chain, and determines whether to also include this data as supporting data **444** relevant to the post-processed prompt **424**. If the context window **434** length of the augmented language model **162** is exceeded, the framework **100** prunes historical session response **442** down until $n=1$. The framework **100** then concatenates the text of the post-processed prompt **424** (after being stripped **423** of filters **422**). Then the framework **100** concatenates the compiled supporting data **444**.

[0085] The process **400** then performs prompt execution **450**. The underlying data architecture platform **140** allows prompts **164** to be run on any type of language model **144**, considering specifics of the language model **144** required for execution, such as context window length. Each user **120** may set the preferred augmented language model **162** for processing of contextualized content **160**. If this is a third-party model, the formed, post-processed prompt **424** (shown as final prompt **454**), including historical interactions/historical session responses **442** and supporting data **444** is sent over an API **172** for processing, and the response **452** of the augmented language model **162** is returned to the user **120** (and/or generated as a defined, domain-specific output **124**). If an open-source, self-hosted language model **144** is selected, it is processed on the internal GPU cluster.

[0086] If the user **120** is executing a task as a defined, domain-specific output **124** which is comprised of a series of sequential actions in an agent task action chain **460**, then the output of a first agent action **462** is piped to the input of the second agent action **464**, and so on, forming a cascading chain reaction of n agent actions **466**. The output of each such agent action in the n agent actions **466** may comprise a defined, domain-specific output **124** of an artificial intelligence-based agent **150**.

[0087] It is to be understood that many types of defined, domain-specific outputs **124** based on contextualized content **160** from private data sets are possible and within the scope of the present invention. Therefore, neither the claims, this specification, nor the present invention, shall be limited to any particular type of domain-specific output **124** or application or use case thereof that is expressed herein.

[0088] The framework **100** generates output data **180** which is utilized to generate the defined, domain-specific output **124**, and there are many exemplary outcomes **190** leveraging this output data **180** to embody the defined, domain-specific output **124**. For example, an artificial intelligence-based agent **150** may generate documents or files **191** that are delivered to external systems **195**. For example, one type of defined, domain-specific output **124** as a document or file **191** may be a short—such as a text, video, or audio file—that may be shared with external systems

195. Examples of external systems **195** in this context include a social media account of a user **120**, or a particular type of external system **195** such as an application that delivers education content, where a short clip of a lecture is desired to be sent to external users. In such an output, the framework **100** automatically generates a file representing a short clip of a portion of a document based on a user instruction in defining a domain-specific output **124**, and exports that file to the destination or system designated by the user **120**.

[0089] Another example of an outcome **190** embodying a defined, domain-specific output **124** is an actuation **193** of an external device or system **193**. This may be accomplished by generating an instruction **194** for such an actuation **193** by an artificial intelligence-based agent **150**. The framework **100** may therefore be configured to analyze information in a private data sets and take action to an external system **195** that automatically actuates some element of that external system **195**—whether it be creating a document or file **191**, or manipulating a mechanical device in some manner (such as for instructing a robot within a laboratory environment). The specific action **192** taken in response to a defined, domain-specific output **124** may therefore be within an external system **195** itself, and the framework **100** may leverage one or more APIs **172** for accomplishing such an outcome.

[0090] One example of an external system **195** is a customer relationship management system **196**. An artificial intelligence-based agent **150** may be configured to execute a specific action **192** in the form of a write function to a customer relationship management system **196** as a defined, domain-specific output **124**.

[0091] It is to be understood that many examples of specific actions **192**, external systems **195**, and defined, domain-specific outputs **124** are possible and within the scope of the present invention, and that many additional applications and use cases **197** are possible. For example, the desired output **124** may be a text string generated in the form of an automatic post to a messaging platform, representing a summary of content the user **120** has designated. In still a further example, the output data **180** representing the defined, domain-specific output **124** may be an instruction **194** to a further large language model (or small language model) to perform a particular task; this may include a string of characters that serves as a natural-language instruction to develop, for example, an advertising campaign, a new curriculum for particular students, an initial draft of an article or book, notes for a song, or any other content, based on the user-designated files **111**. In a more specific example of this, a user **120** performs a search using the artificial intelligence-based agent **150** of the present invention to locate and summarize all instances within the user-designated files **111** where a new idea for an invention was discussed. The framework **100** may summarize all of those instances and generate an instruction for an augmented language model **162** to prepare an initial draft of a patent application for the invention that was discussed, and return the draft to the user **120**.

[0092] One example of an artificial intelligence-based agent **150** is an Auto-Grade agent. In an Auto-Grade agent, user-designated files **111** may include question text, teacher-provided answer key, text of student answers to each question, and related source educational material as input data **110**. If the source material includes media files such as audio files **114** or video files **115**, the framework **100** first transcribes those to transform the source material to text. A defined, domain-specific output **124** of the Auto-Grade agent in this example may include grade, and justification of the grade, to each student's answer to each question.

[0093] In this example, each set of questions are typically related to some source material, such as a textbook chapter or video. These data sources are often proprietary and likely not provided in the generalized training data for the language model **144** being used. For each question, the artificial intelligence-based agent **150** forms a prompt **164** that includes the question, the answer, the teacher-provided answer key, grading output constraints (e.g., grade must be between 0 and 1 inclusive, partial marks are allowed), instructions for how to provide justifications for provided grades, and include any relevant supporting data from the related educational material. If student

answers are being batch processed (as human teachers often do), the artificial intelligence-based agent **150** packs as many student answers into one prompt as possible, asking the augmented language model **162** to provide a grade and justification to each student answer in a structured format such as JSON. This batching technique is predominantly for speed and economic efficiency. [0094] Packing all of this data into a prompt **164** requires balancing because of the limited context window length—a prompt **164** for a language model can only take in as input a fixed number of tokens. If supporting data needs to be compressed to perform this balancing, RAG techniques such as cosine similarity **417** are implemented with the prompt embedding vector to prune **436** irrelevant data. The artificial intelligence-based agent **150** may also decrease the batch size (number of student responses being processed at once) to ensure for highest efficiency with the limited context window availability.

[0095] There are many types of machine learning techniques that may be implemented within the supporting data architecture platform **140** and the framework **100**. Graph data models are one example of machine learning tools **142**. Graph data models leverage a variety of machine learning algorithms to enhance their functionality in knowledge graph implementations involving artificial intelligence models. These algorithms assist in uncovering insights and patterns within interconnected data to enhance performance of artificial intelligence models such as language models **162**. Machine learning algorithms may be integrated with knowledge graphs for tasks such as entity classification, sentiment analysis, and predictive modeling. Some specific machine learning-based algorithms that may be used include link prediction algorithms, such as node embedding methods, are used to predict missing relationships or edges between entities in the knowledge graph. These algorithms learn representations of nodes (entities) in a way that encodes their structural and semantic properties, enabling the identification of potential connections that may not be explicitly present in the data.

[0096] Graph traversal algorithms such as Breadth-First Search (BFS) and Depth-First Search (DFS) may be utilized for graph traversal. These types of algorithms help to navigate the knowledge graph to discover connected entities, uncovering paths and relationships that provide valuable insights into the data. Community detection algorithms, such as Louvain or Modularity-based methods, may also be applied to identify densely-connected clusters or communities within the knowledge graph. These communities often represent groups of related entities or topics, aiding in the organization and categorization of knowledge when applied to an output of a language model **162** to improve the accuracy of such an output. Centrality algorithms may also be used to determine the importance or influence of nodes within the graph. These may be used to identify key entities or hubs in the knowledge graph, which may be valuable for recommendation systems or understanding the significance of different data points. Algorithms specific to semantic similarity and natural language processing (NLP) techniques enable word embeddings to be integrated with knowledge graphs to capture semantic relationships between entities and textual information. This enables the extraction of knowledge from unstructured text data and its integration into the graph.

[0097] Knowledge graph embeddings algorithms may be used to transform entities and relationships into continuous vector representations. These embeddings enable mathematical operations on the graph structure and support various tasks of AI tools such as language models **162**, such as entity classification, relation prediction, and question answering. Other algorithms for specifically improving augmentations of language models **162** include graph neural networks (GNNs). GNNs are deep learning models designed for graph-structured data. They leverage message passing and aggregation mechanisms to perform tasks like node classification, link prediction, and graph classification. GNNs are examples of tools for linking knowledge graph models and large language models, by incorporating both structural and attribute information from the knowledge graph into the AI models **160**.

[0098] Machine learning tools **142** in the framework **100** of the present invention may also include,

as noted above, applications of neural networks. Neural networks generally are comprised of nodes, which are computational units having one or more biased input/output connections. Such biased connections act as transfer (or activation) functions that combine inputs and outputs in some way. Nodes are organized into multiple layers that form the neural network. There are many types of neural networks, which are computing systems that “learn” to perform tasks in a supervised manner, without being programmed with task-specific rules, based on examples.

[0099] Neural networks generally are based on arrays of connected, aggregated nodes (or, “neurons”) that transmit signals to each other in the multiple layers over the biased input/output connections. Connections, as noted above, are activation or transfer functions which “fire” these nodes and combine inputs according to mathematical equations or formulas. Different types of neural networks generally have different configurations of these layers of connected, aggregated nodes, but they can generally be described as an input layer, a middle or ‘hidden’ layer, and an output layer. These layers may perform different transformations on their various inputs, using different mathematical calculations or functions. Signals travel between these layers, from the input layer to the output layer via the middle layer, and may traverse layers, and nodes, multiple times.

[0100] Signals are transmitted between nodes over connections, and the output of each node is calculated in a non-linear function that sums all of the inputs to that node. Weight matrices and biases are typically applied to each node, and each connection, and these weights and biases are adjusted as the neural network processes inputs and transmits them across the nodes and connections. These weights represent increases or decreases in the strength of a signal at a particular connection. Additionally, nodes may have a threshold, such that a signal is sent only if the aggregated output at that node crosses that threshold. Weights generally represent how long an activation function takes, while biases represent when, in time, such a function starts; together, they help gradients minimize over time. At least in the case of weights, they can be initialized and change (i.e., decay) over time, as a system learns what weights should be, and how they should be adjusted. In other words, neural networks evolve as they learn, and the mathematical formulas and functions that comprise neural networks design can change over time as a system improves itself.

[0101] Neural networks can also incorporate a time delay, or feedback loop, which is calculated to generally account for temporal dependencies, to further improve the results of the modeling framework. This may be used by a particular type of neural network that accounts for timed data sequences, such as for example the Long-Short-Term-Memory (LSTM) neural network, discussed above. Feedback loops and other time delay mechanisms applied by the various mathematical functions of such a neural network are modeled after one or more temporally-relevant characteristics, and incorporate calculated weights and biases for variables depending on the input data collected and type of problem being analyzed.

[0102] Neural networks may be configured to address the problem of decay in longer time-dependent sequences in an architecture that has multiple, interactive components acting as “blocks” in place of the conventional layers of the neural network. Each of these blocks may represent a single layer in a middle layer, or may form multiple layers; regardless, each block may be thought of as representing different timesteps in a time-dependent sequence analysis of input data.

[0103] The components of such a specially-focused neural network form its internal state and include a cell, which acts as the memory portion of the block, and three regulating gates that control the flow of information inside each block: an input gate, an output gate, and a forget gate. The cell remembers values over arbitrary time intervals, by keeping track of the dependencies between elements in an input sequence, and the three gates regulate the flow of information into and out of the cell. The input gate controls the extent to which a new value flows into the cell, the forget gate controls the extent to which a value remains in the cell, and the output gate controls the extent to which the value in the cell is used to compute the output of the block. The decision-making function of these gates is often referred to as the logistic sigmoid function for computing outputs of gates in these types of neural networks. There are connections into and out of these

gates, and at least the weights of these connections, which need to be learned during training, determine how the gates operate.

[0104] Inside neural network blocks, there are additional layers that perform the activation functions needed to ensure that time-dependent data sequences are properly analyzed to avoid decay. One such activation function that may be incorporated is a tan h layer, which effectively classifies input data by determining which input values are added to the internal state of the block. Input gates are a layer of sigmoid-activated nodes whose output is multiplied by inputs classified by preceding tan h layers. The effect of these activation functions is to filter any elements of the inputs that are not required, based on the values assigned to each node for the problem being analyzed, and the weights and biases applied. The weights applied to connections between these nodes can be trained to output values close to zero to switch off certain input values (or, conversely, to pass through other values). Another internal state of a block, the forget gate, is effectively a feedback loop that operates to create a layer of recurrence that further reduces the risk of decay in time-dependent input data. The forget gate helps the neural network learn which state variables should be “remembered” or “forgotten”.

[0105] Supervised learning is an application of mathematical functions in algorithms that classify input data to find specific relationships or structure therein that allow the machine learning prediction engine to efficiently produce highly accurate output data. There are many types of such algorithms for performing mathematical functions in supervised learning approaches. These include regression analysis (including the logistic regression discussed above, and polynomial regression, and many others), decision trees, Bayesian approaches such as naive Bayes, support vector machines, random forests, anomaly detection, etc.

[0106] Recurrent neural networks are a name given to types of neural networks in which connections between nodes follow a directed temporal sequence, allowing the neural network to model temporal dynamic behavior and process sequences of inputs of variable length. These types of neural networks are deployed where there is a need for recognizing, and/or acting on, such sequences. As with neural networks generally, there are many types of recurrent neural networks.

[0107] Neural networks having a recurrent architecture may also have stored, or controlled, internal states which permit storage under direct control of the neural network, making them more suitable for inputs having a temporal nature. This storage may be in the form of connections or gates which act as time delays or feedback loops that permit a node or connection to retain data that is prior in time for modeling such temporal dynamic behavior. Such controlled internal states are referred to as gated states or gated memory, and are part of long short-term memory networks (LSTMs) and gated recurrent units (GRUs), which are names of different types of recurrent neural network architectures. This type of neural network design is utilized where desired outputs of a system are motivated by the need for memory, as storage, and as noted above, where the system is designed for processing inputs that are comprised of timed data sequences. Examples of such timed data sequences include video, speech recognition, and handwriting—where processing requires an analysis of data that changes temporally. In the present invention, where output data is in the form of predicted or forecasted future state of some condition, an understanding of the influence of various events on a state over a period of time lead to more highly accurate and reliable predictions or forecasts.

[0108] Many other types of recurrent neural networks exist. These include, for example, fully recurrent neural networks, Hopfield networks, bi-directional associative memory networks, echo state networks, neural Turing machines, and many others, all of which exhibit the ability to model temporal dynamic behavior. Any instantiation of such neural networks in the present invention may include one or more of these types, and it is to be understood that neural networks applied within the machine learning prediction engine may include different ones of such types. Therefore, the present invention contemplates that many types of neural networks may be implemented, depending at least on the type of problem being analyzed.

[0109] The systems and methods of the present invention may be implemented in many different computing environments. For example, the various algorithms embodied in the data processing elements may each be implemented in conjunction with a special purpose computer, a programmed microprocessor or microcontroller and peripheral integrated circuit element(s), an ASIC or other integrated circuit, a digital signal processor, electronic or logic circuitry such as discrete element circuit, a programmable logic device or gate array such as a PLD, PLA, FPGA, PAL, and any comparable means. In general, any means of implementing the methodology illustrated herein can be used to implement the various aspects of the present invention. Exemplary hardware that can be used for the present invention includes computers, handheld devices, telephones (e.g., cellular, Internet enabled, digital, analog, hybrids, and others), and other such hardware. Some of these devices include processors (e.g., a single or multiple microprocessors), memory, nonvolatile storage, input devices, and output devices. Furthermore, alternative software implementations including, but not limited to, distributed processing, parallel processing, or virtual machine processing can also be configured to perform the methods described herein.

[0110] The systems and methods of the present invention may also be partially implemented in software that can be stored on a storage medium, executed on a programmed general-purpose computer with the cooperation of a controller and memory, a special purpose computer, a microprocessor, or the like. In these instances, the systems and methods of this invention can be implemented as a program embedded on personal computer such as an applet, JAVA® or CGI script, as a resource residing on a server or computer workstation, as a routine embedded in a dedicated measurement system, system component, or the like. The system can also be implemented by physically incorporating the system and/or method into a software and/or hardware system.

[0111] Additionally, the data processing functions disclosed herein may be performed by one or more program instructions stored in or executed by such memory, and further may be performed by one or more modules configured to carry out those program instructions. Modules are intended to refer to any known or later developed hardware, software, firmware, artificial intelligence, fuzzy logic, expert system or combination of hardware and software that is capable of performing the data processing functionality described herein.

[0112] The foregoing descriptions of embodiments of the present invention have been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Accordingly, many alterations, modifications and variations are possible in light of the above teachings, may be made by those having ordinary skill in the art without departing from the spirit and scope of the invention. For example, it is to be understood that artificial intelligence-based agents **150** may run natively within the framework **100** and supporting data architecture platform **140**, or within external systems **195**, or within a cloud computing-based container running a particular workload or process. Artificial intelligence-based agents **150** may therefore be embedded either within the supporting data architecture **140**, within a cloud computing environment, or within an external device itself. It is therefore intended that the scope of the invention be limited not by this detailed description. For example, notwithstanding the fact that the elements of a claim are set forth below in a certain combination, it must be expressly understood that the invention includes other combinations of fewer, more or different elements, which are disclosed in above even when not initially claimed in such combinations.

[0113] The words used in this specification to describe the invention and its various embodiments are to be understood not only in the sense of their commonly defined meanings, but to include by special definition in this specification structure, material or acts beyond the scope of the commonly defined meanings. Thus if an element can be understood in the context of this specification as including more than one meaning, then its use in a claim must be understood as being generic to all possible meanings supported by the specification and by the word itself.

[0114] The definitions of the words or elements of the following claims are, therefore, defined in

this specification to include not only the combination of elements which are literally set forth, but all equivalent structure, material or acts for performing substantially the same function in substantially the same way to obtain substantially the same result. In this sense it is therefore contemplated that an equivalent substitution of two or more elements may be made for any one of the elements in the claims below or that a single element may be substituted for two or more elements in a claim. Although elements may be described above as acting in certain combinations and even initially claimed as such, it is to be expressly understood that one or more elements from a claimed combination can in some cases be excised from the combination and that the claimed combination may be directed to a sub-combination or variation of a sub-combination.

[0115] Insubstantial changes from the claimed subject matter as viewed by a person with ordinary skill in the art, now known or later devised, are expressly contemplated as being equivalently within the scope of the claims. Therefore, obvious substitutions now or later known to one with ordinary skill in the art are defined to be within the scope of the defined elements.

[0116] The claims are thus to be understood to include what is specifically illustrated and described above, what is conceptually equivalent, what can be obviously substituted and also what essentially incorporates the essential idea of the invention.

Claims

1. A method, comprising: receiving input data comprised of one or more files representing a set of information for a defined, domain-specific output of an artificial intelligence-based agent; analyzing the one or more files in response to at least one prompt of artificial intelligence-based agent to generate the defined, domain-specific output, by: preparing contextualized content from the set of information in a plurality of machine learning tools within a supporting architecture platform for the artificial intelligence-based agent, augmenting a selected language model with the contextualized content, and prompting the selected language model within the artificial intelligence-based agent with one or more prompts, wherein the artificial intelligence-based agent generates the defined, domain-specific output based on the augmented selected language model and the contextualized content; and wherein the artificial intelligence-based agent executes a specific action identified by the user external to the supporting architecture platform from the defined, domain-specific output.
2. The method of claim 1, further comprising providing one or more instructions to perform the specific action identified by the user to an external system.
3. The method of claim 2, wherein the specific action is executed within the external system.
4. The method of claim 2, wherein the external system is a customer relationship management system, and wherein the specific action executed is a write function of output data comprised of the defined, domain-specific output to the customer relationship management system.
5. The method of claim 1, wherein the artificial intelligence-based agent is embedded in an external system and executes the specific action within the external system.
6. The method of claim 1, wherein the specific action identified by the user external to the supporting architecture platform is an actuation of an external device.
7. The method of claim 1, wherein the one or more files comprise private data sets designated by the user, and wherein the private data sets define contextual and temporal constraints that are domain-specific for the defined, domain-specific output of the artificial intelligence-based agent.
8. The method of claim 1, wherein the plurality of native machine learning tools are configured to prepare the content in the one or more files by extracting data points from unstructured portions of the content that are relative to the defined, domain-specific output, creating individual embeddings from the data points, and calculating embedding vectors for the content.
9. The method of claim 1, further comprising implementing a retrieval-augmented generation architecture to extract information relative to the defined, domain-specific output from the one or

more files, and adding information extracted by the retrieval-augmented generating architecture to the contextualized content where the set of information for the defined, domain-specific output includes one or more structured portions.

10. The method of claim 1, wherein the prompting a selected language model within the artificial intelligence-based agent with one or more prompts further comprises providing one or more verbal, written, or gestured prompts of the artificial intelligence-based agent by the user.

11. The method of claim 1, further comprising enabling a user to define a domain-specific output of the artificial intelligence-based agent for the one or more files.

12. The method of claim 1, further comprising selecting a language model, wherein the language model is selected either by a user or automatically by the artificial intelligence-based agent.

13. The method of claim 1, wherein the prompting the selected language model within the artificial intelligence-based agent with one or more prompts includes nested sub-prompts for the selected language model that are responsive to user changes to one or both of the defined, domain-specific output and the specific action.

14. A method, comprising: contextualizing content in one or more user-designated files representing a set of information for a defined, domain-specific output of an artificial intelligence-based agent in a plurality of machine learning tools; augmenting a selected language model with the contextualized content; and analyzing the contextualized content within the artificial intelligence-based agent, by prompting the selected language model with one or more prompts, wherein the artificial intelligence-based agent generates the defined, domain-specific output based on the augmented selected language model and the contextualized content, and wherein the defined, domain-specific output of the artificial intelligence-based agent executes a specific action identified by the user external to a data architecture platform supporting the artificial intelligence-based agent.

15. The method of claim 14, wherein the defined, domain-specific output from the artificial intelligence-based agent includes one or more instructions provided to an external system to execute the specific action.

16. The method of claim 15, wherein the specific action is executed within the external system.

17. The method of claim 15, wherein the external system is a customer relationship management system, and wherein the specific action executed is a write function of output data comprised of the defined, domain-specific output to the customer relationship management system.

18. The method of claim 14, wherein the artificial intelligence-based agent is embedded in an external system and executes the specific action within the external system.

19. The method of claim 14, wherein the specific action identified by the user external to the supporting architecture platform is an actuation of an external device.

20. The method of claim 14, wherein the one or more files comprise private data sets designated by the user, and wherein the private data sets define contextual and temporal constraints that are domain-specific for the defined, domain-specific output of the artificial intelligence-based agent.

21. The method of claim 14, wherein the plurality of native machine learning tools are configured to prepare the content in the one or more files by extracting data points from unstructured portions of the content that are relative to the defined, domain-specific output, creating individual embeddings from the data points, and calculating embedding vectors for the content.

22. The method of claim 14, further comprising implementing a retrieval-augmented generation architecture to extract information relative to the defined, domain-specific output from the one or more files, and adding information extracted by the retrieval-augmented generating architecture to the contextualized content where the set of information for the defined, domain-specific output includes one or more structured portions.

23. The method of claim 14, wherein the prompting a selected language model within the artificial intelligence-based agent with one or more prompts further comprises providing one or more verbal, written, or gestured prompts of the artificial intelligence-based agent by the user.

24. The method of claim 14, further comprising enabling a user to define a domain-specific output of the artificial intelligence-based agent for the one or more files.
25. The method of claim 14, further comprising selecting a language model, wherein the language model is selected either by a user or automatically by the artificial intelligence-based agent.
26. The method of claim 14, wherein the prompting the selected language model within the artificial intelligence-based agent with one or more prompts includes nested sub-prompts for the selected language model that are responsive to user changes to one or both of the defined, domain-specific output and the specific action.
27. A system, comprising: one or more artificial intelligence-based agents within a supporting data architecture platform, the one or more artificial intelligence-based agents configured to: contextualize content in one or more user-designated files representing a set of information for a defined, domain-specific output of the artificial intelligence-based agent in a plurality of machine learning tools, augment a selected language model with the contextualized content, and analyze the contextualized content within the artificial intelligence-based agent, by prompting the selected language model with one or more prompts, wherein the artificial intelligence-based agent generates the defined, domain-specific output based on the augmented selected language model and the contextualized content, and wherein the defined, domain-specific output of the artificial intelligence-based agent executes a specific action identified by the user external to a data architecture platform supporting the artificial intelligence-based agent.
28. The system of claim 27, wherein the defined, domain-specific output from the artificial intelligence-based agent includes one or more instructions provided to an external system to execute the specific action.
29. The system of claim 28, wherein the specific action is executed within the external system.
30. The system of claim 28, wherein the external system is a customer relationship management system, and wherein the specific action executed is a write function of output data comprised of the defined, domain-specific output to the customer relationship management system.
31. The system of claim 27, wherein the artificial intelligence-based agent is embedded in an external system and executes the specific action within the external system.
32. The system of claim 27, wherein the specific action identified by the user external to the supporting architecture platform is an actuation of an external device.
33. The system of claim 27, wherein the one or more files comprise private data sets designated by the user, and wherein the private data sets define contextual and temporal constraints that are domain-specific for the defined, domain-specific output of the artificial intelligence-based agent.
34. The system of claim 27, wherein the plurality of native machine learning tools are configured to prepare the content in the one or more files by extracting data points from unstructured portions of the content that are relative to the defined, domain-specific output, creating individual embeddings from the data points, and calculating embedding vectors for the content.
35. The system of claim 27, further comprising implementing a retrieval-augmented generation architecture to extract information relative to the defined, domain-specific output from the one or more files, and adding information extracted by the retrieval-augmented generating architecture to the contextualized content where the set of information for the defined, domain-specific output includes one or more structured portions.
36. The system of claim 27, wherein the prompting a selected language model within the artificial intelligence-based agent with one or more prompts further comprises providing one or more verbal, written, or gestured prompts of the artificial intelligence-based agent by the user.
37. The system of claim 27, further comprising enabling a user to define a domain-specific output of the artificial intelligence-based agent for the one or more files.
38. The system of claim 27, further comprising selecting a language model, wherein the language model is selected either by a user or automatically by the artificial intelligence-based agent.
39. The method of claim 27, wherein the prompting the selected language model within the

artificial intelligence-based agent with one or more prompts includes nested sub-prompts for the selected language model that are responsive to user changes to one or both of the defined, domain-specific output and the specific action.
